ADVANCED DATABASE ADMINISTRATION

- ➢Database User Security
- ➢Managing Schema Objects
- ➢Managing Data And Concurrency

UNIT-2

By Jasleen Kaur
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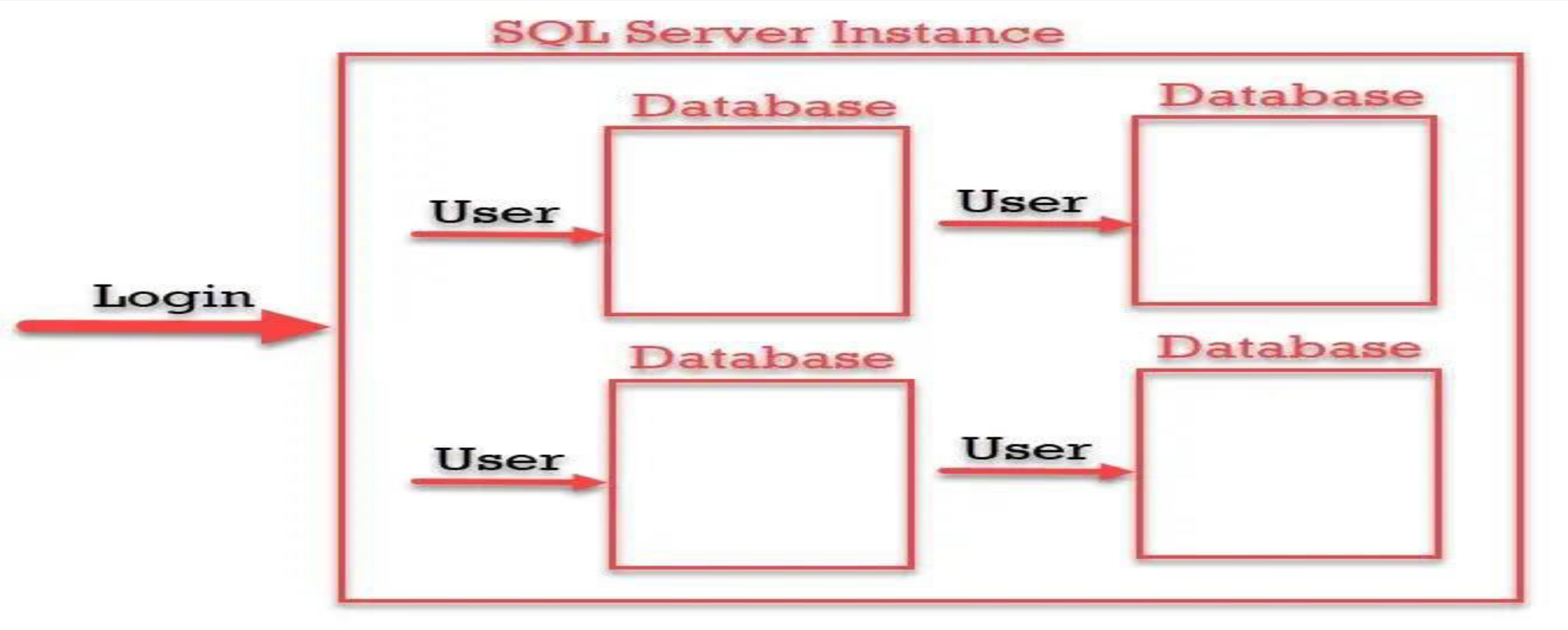
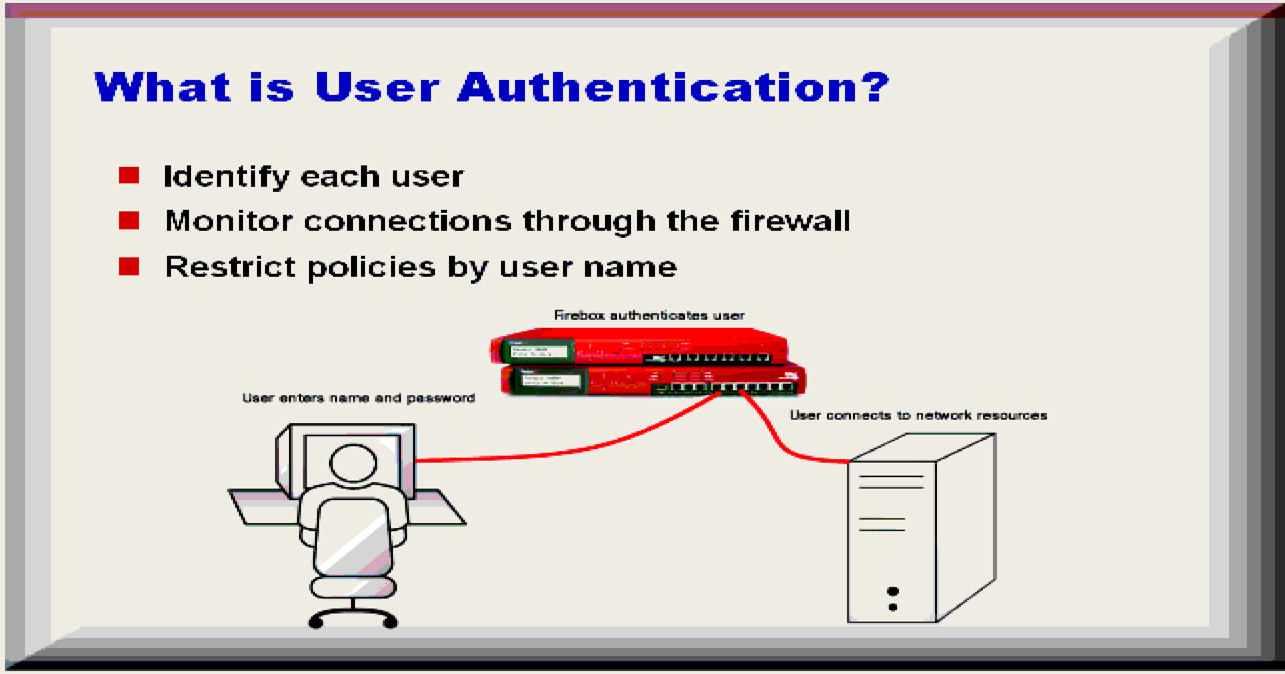
DATABASE USER SECURITY

- **Creating a New User Account-** You create a database user with the CREATE USER statement
- Because it is a powerful privilege, a database administrator or security administrator is usually the only user who has the CREATE USER system privilege.
- Creates a user and specifies the user password, default tablespace, temporary tablespace where temporary segments are created, tablespace quotas, and profile



Authenticating User

- Authentication means verifying the identity of someone (a user, device, or other entity) who wants to use data, resources, or applications.
- Validating that identity establishes a trust relationship for further interactions
- Authentication also enables accountability by making it possible to link access and actions to specific identities.
- After authentication, authorization processes can allow or limit the levels of access and action permitted to that entity.
- Oracle Database requires special authentication procedures for database administrators, because they perform special database operations.
- Oracle Database also encrypts passwords during transmission to ensure the security of network authentication.



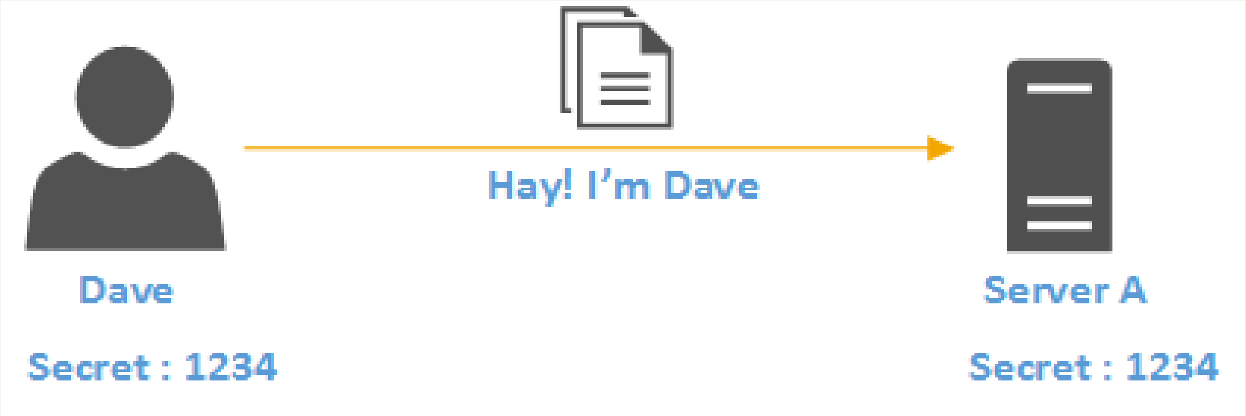
How user authentication works

- In order to gain access, users must prove to the website that they are who they say they are.
- The ID and key are enough to confirm the user's identity, which will allow the system to authorize the user.
- It's important to note that authorization, on the other hand, is what dictates what users are able to see and do when they log in.
- While authorization and authentication are often used interchangeably, the two *different* terms work together to create a secure login process.
- **To put it simply, user authentication has three tasks:**

1. Manage the connection between the human (user) and the website's server (computer).
2. Verify users' identities.
3. Approve (or decline) the authentication so the system can move to authorizing the user.

- The process is fairly simple; users input their credentials on the website's login form. That information is then sent to the authentication server where the information is compared with all the user credentials on file.
- When a match is found, the system will authenticate users and grant them access to their accounts.

- If a match isn't found, users will be prompted to re-enter their credentials and try again.
- After several unsuccessful attempts, the account may be flagged for suspicious activity or require alternative authentication methods such as a password reset or a one time password.



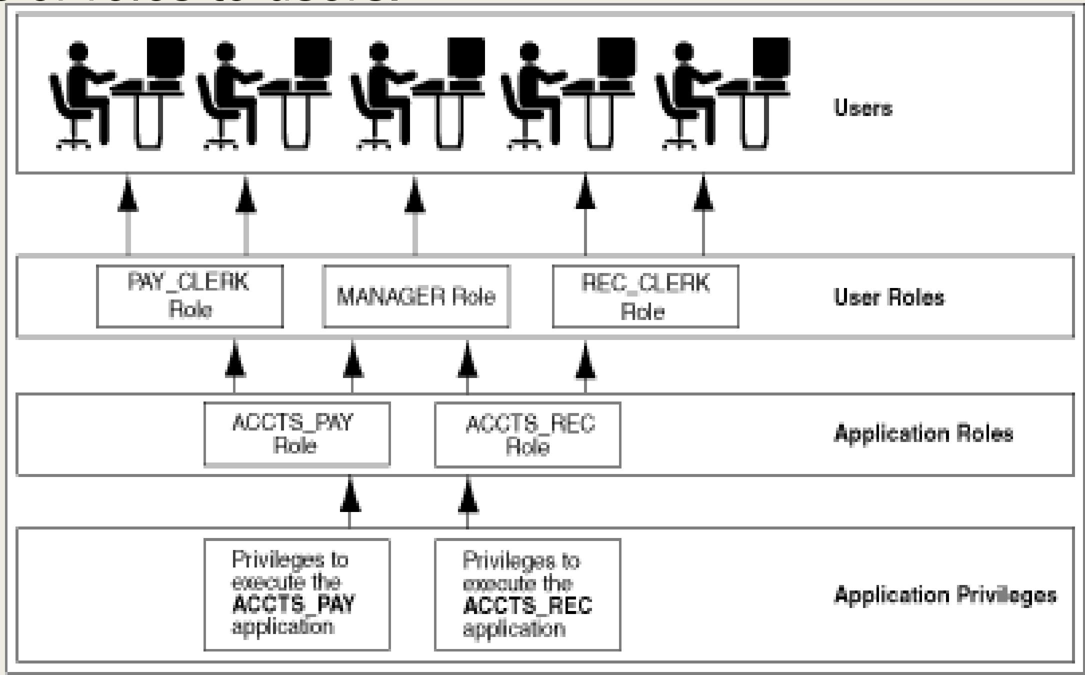
Privileges and Roles

- Authorization includes primarily two processes:

☐ Permitting only certain users to access, process, or alter data.

☐ Applying varying limitations on user access or actions. The limitations placed on users can apply to objects such as schemas, tables, or rows or to resources such as time.

- A **user privilege** is the right to run a particular type of SQL statement, or the right to access an object that belongs to another user, run a PL/SQL package.
- **Roles** are created by users (usually administrators) to group together privileges or other roles. They are a way to facilitate the granting of multiple privileges or roles to users.
- **User roles** groups several privileges and roles, so that they can be granted to and revoked from users simultaneously. You must enable the role for a user before the user can use it.



➢ The types of privileges are defined by Oracle Database:

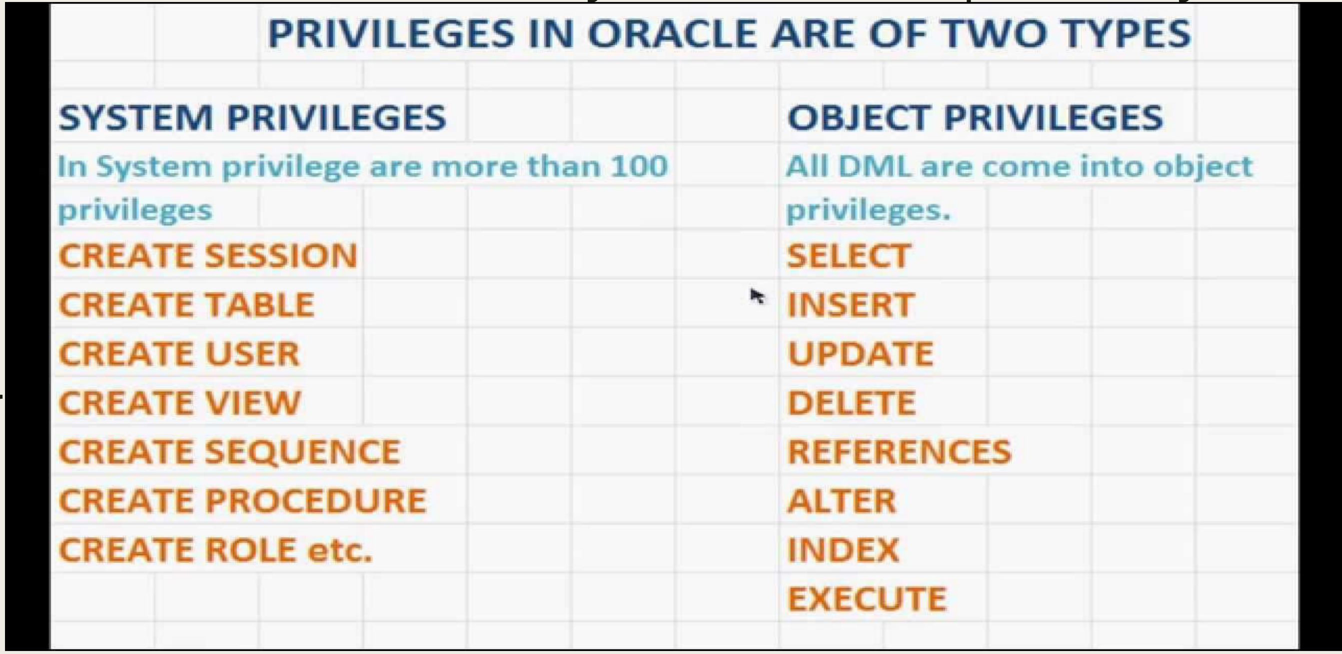
- **System privileges-** These privileges allow to perform standard administrator tasks in the database. Restrict them only to trusted users. For example, the privileges to create tablespaces and to delete the rows of any table in a database are system privileges.

➢ Because system privileges are so powerful, by default the data base is configured to prevent typical(non-administrative) users from exercising the ANY system privileges (such as UPDATE ANY TABLE) on the data dictionary

- **Object privileges-** An object privilege is a right that you grant to a user on a database object. Each type of object has privileges associated with it.

➢ An object privilege is a right that you grant to a user on a database object. Some examples of object privileges include the right to:

- Use an edition
- Update a table
- Select rows from another user's table
- Execute a stored procedure of another user


PRIVILEGES IN ORACLE ARE OF TWO TYPES

SYSTEM PRIVILEGES	OBJECT PRIVILEGES
In System privilege are more than 100 privileges	All DML are come into object privileges.
CREATE SESSION	SELECT
CREATE TABLE	INSERT
CREATE USER	UPDATE
CREATE VIEW	DELETE
CREATE SEQUENCE	REFERENCES
CREATE PROCEDURE	ALTER
CREATE ROLE etc.	INDEX
	EXECUTE

Granting Object Privileges

- ☐ Grants the SELECT, INSERT, and DELETE object privileges for all columns of the emp table to the user jfee.

```
GRANT SELECT, INSERT, DELETE ON emp TO jfee;
```

- ☐ Grant all object privileges on the salary view to user jfee, use the ALL keyword as shown in the following example:

```
GRANT ALL ON salary TO jfee;
```

MANAGING SCHEMA OBJECT

As a database administrator (DBA), you can create, modify, and delete schema objects in your own schema and in any other schema. A database administrator is defined as any user who is granted the DBA role.

Table Types

- The table is the basic unit of data storage in an Oracle database. It holds all user accessible data. Each table is made up of columns and rows.
- The most common type of table in an Oracle database is a RELATIONAL TABLE, which is structured with simple columns. Two other table types are supported: OBJECT TABLES and XML TYPE TABLES. Any of the three table types can be defined as *permanent* or *temporary.*
- Temporary tables hold session-private data that exists only for the duration of a transaction or session. They are useful in applications where a results set must be held temporarily in memory
- You can build relational tables in either *heap* or *index-organized* structures.
- In heap structures, the rows are not stored in any particular order.
- In index-organized tables, the row order is determined by the values in one or more selected columns

Actions with Table

- <u>Creating Tables</u>: To create a new table in your schema, you must have the CREATE TABLE system privilege. To create a table in another user's schema, you must have the CREATE ANY TABLE system privilege.

  CREATE TABLE hr.admin_emp(

  empno NUMBER(5) PRIMARY KEY,

  ename VARCHAR2(15) NOT NULL,

- <u>Adding Table Columns</u>:

  ALTER TABLE hr.admin_emp

  ADD (bonus NUMBER(7,2));

- <u>Renaming Table Columns</u>:

  ALTER TABLE hr.admin_emp

  RENAME COLUMN comm TO commission;

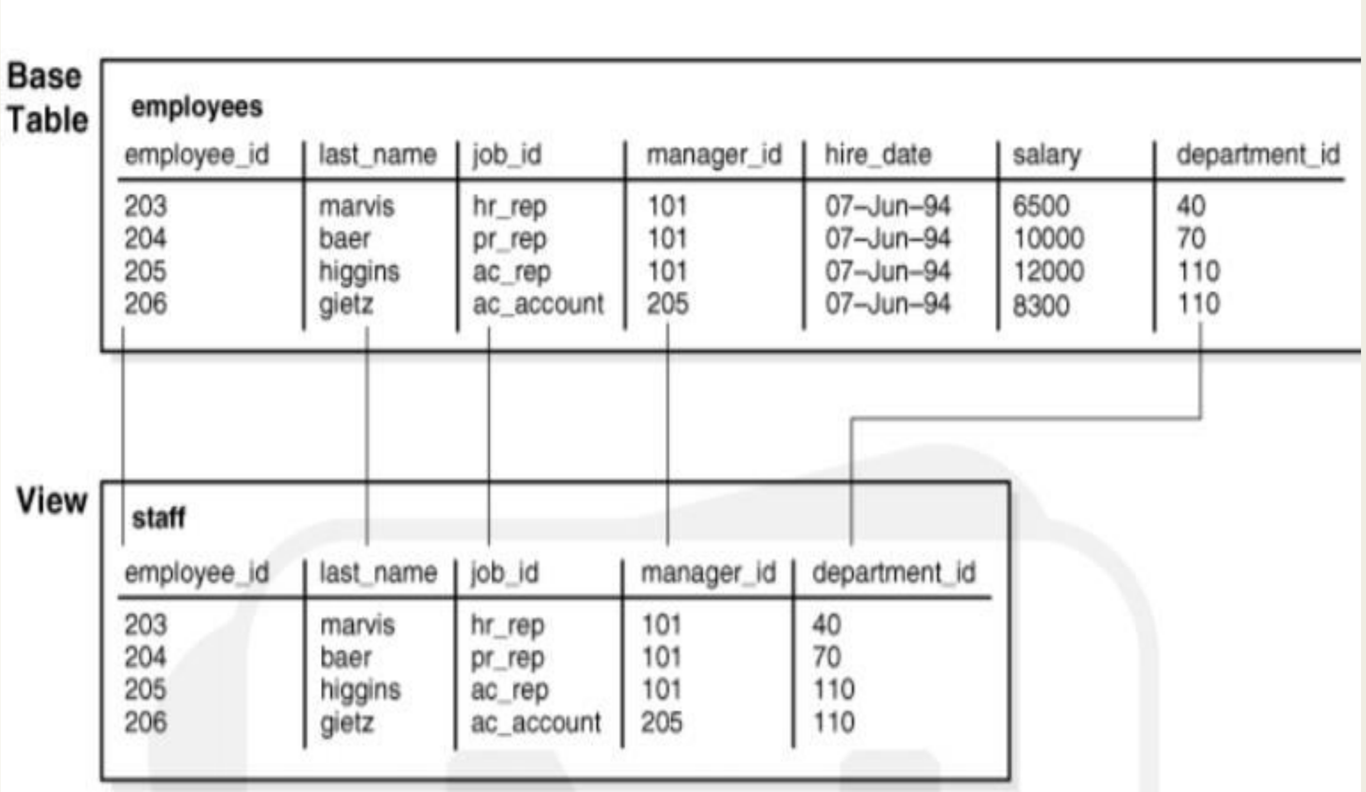
- <u>Dropping Table Columns:</u>

  ALTER TABLE hr.admin_emp DROP COLUMN sal;

  ALTER TABLE hr.admin_emp DROP (bonus, commission);

Views

- A view is a tailored presentation of the data contained in one or more tables or other views. A view takes the output of a query and treats it as a table.
- Therefore, a view can be thought of as a **stored query** or a **virtual table**. You can use views in most places where a table can be used
- For example, the employees table has several columns and numerous rows of information. If you want users to see only five of these columns or only specific rows, then you can create a view of that table for other users to access.



Sequence Generator

- The sequence generator provides a sequential series of numbers. The sequence generator is especially useful in multiuser environments for generating unique sequential numbers
- For example, assume two users are simultaneously inserting new employee rows into the employees table. By using a sequence to generate unique employee numbers for the employee_id column, neither user has to wait for the other to enter the next available employee number. The sequence automatically generates the correct values for each user.
- A sequence definition indicates general information, such as the following:

☐ The name of the sequence

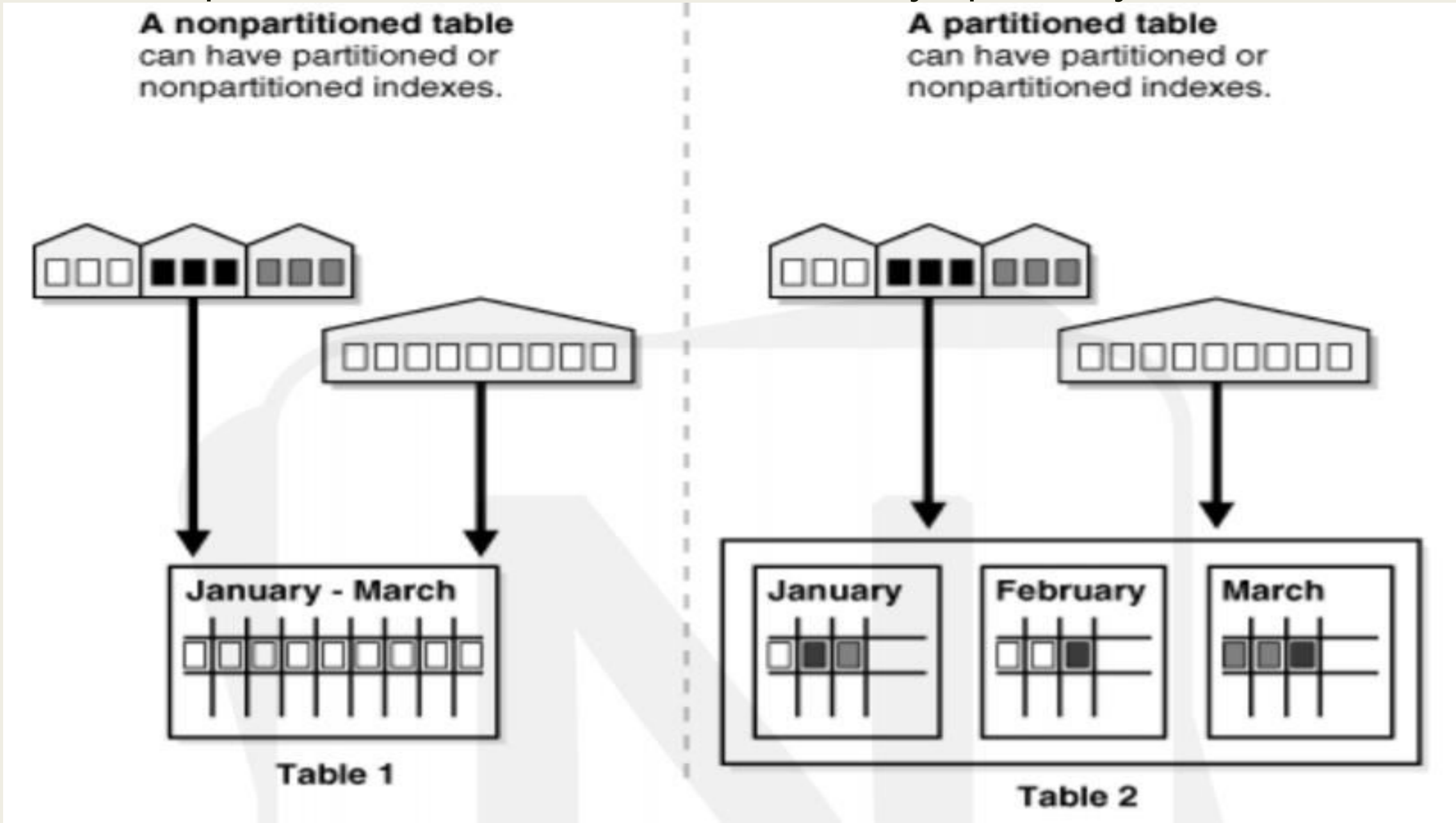
☐ Whether the sequence ascends or descends

☐ The interval between numbers

☐ Whether Oracle should cache sets of generated sequence numbers in memory

Partitioning

- Partitioning enhances the performance, manageability, and availability of a wide variety of applications and helps reduce the total cost of ownership for storing large amounts of data.
- Partitioning allows tables, indexes, and index-organized tables to be subdivided into smaller pieces, enabling these database objects to be managed and accessed at a finer level of granularity.
- Each partition has its own name, and may optionally have its own storage characteristics.



- Benefits of Partitioning

- Improve performance, manageability, and availability

- Partitioning can greatly simplify common administration tasks.

- Partitioning also enables data base designers and administrators to tackle some of the toughest problems

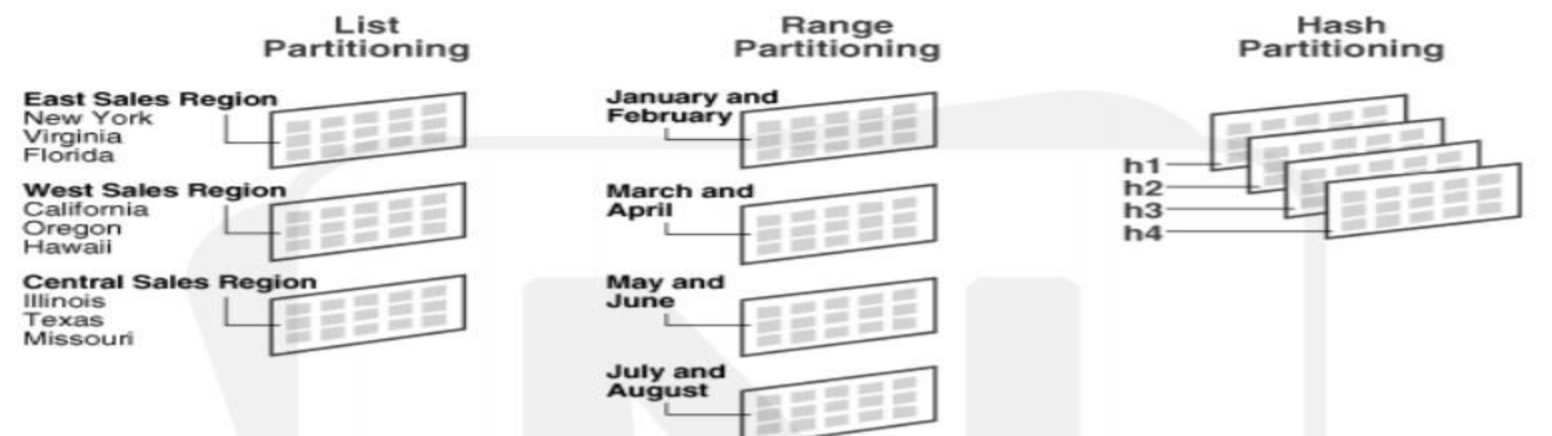
Partitioning Strategies

Levels:

- Single-Level Partitioning
- Composite Partitioning

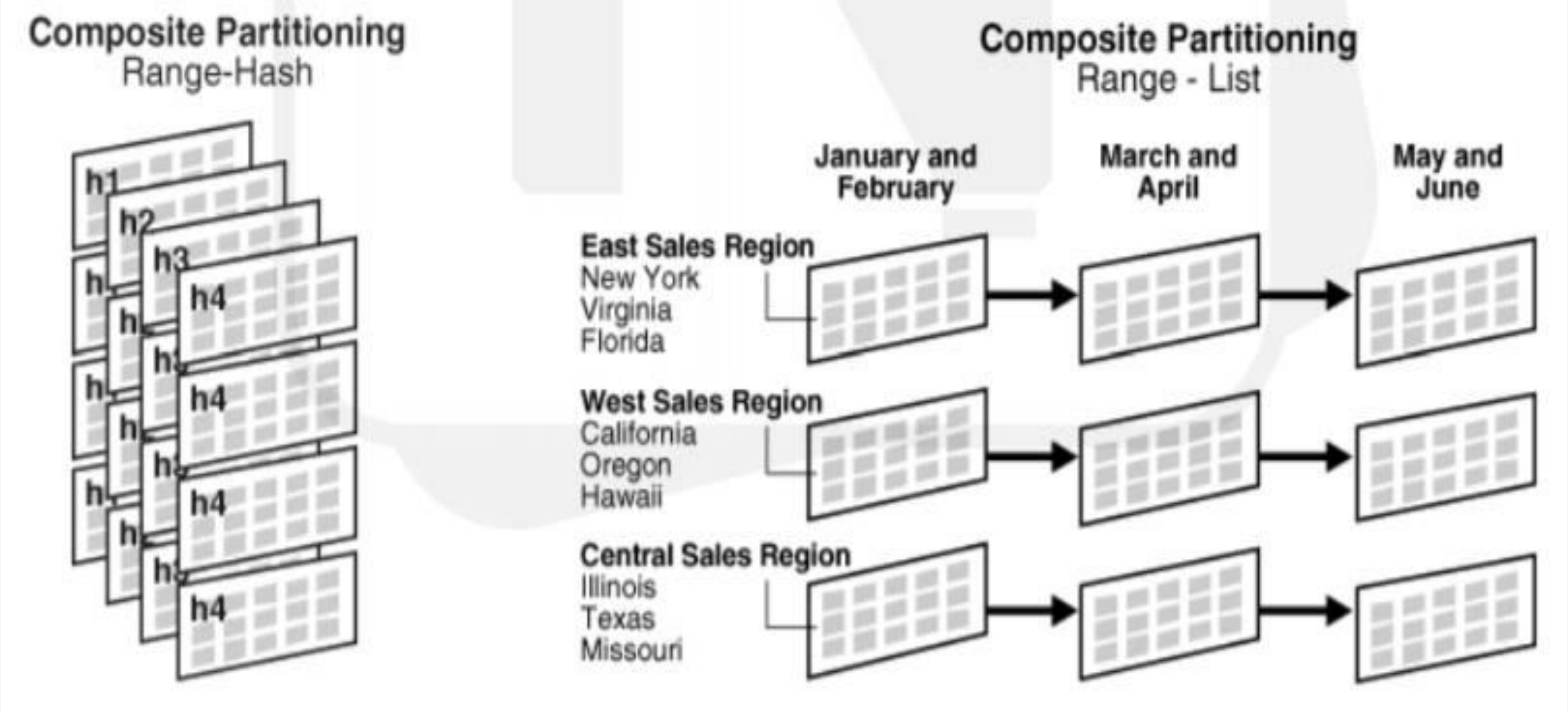
Single-Level Partitioning:

- Range Partitioning: Range partitioning maps data to partitions based on ranges of values of the partitioning key that you establish for each partition
- Hash Partitioning: Hash partitioning maps data to partitions based on a hashing algorithm that Oracle applies to the partitioning key that you identify. The hashing algorithm evenly distributes rows among partitions, giving partitions approximately the same size.
- List Partitioning: List partitioning enables you to explicitly control how rows map to partitions by specifying a list of discrete values for the partitioning key in the description for each partition. The advantage of list partitioning is that you can group and organize unordered and unrelated sets of data in a natural way.



Composite Partitioning

Composite partitioning is a combination of the basic data distribution methods; a table is partitioned by one data distribution method and then each partition is further subdivided into subpartitions using a second data distribution method. All subpartitions for a given partition together represent a logical subset of the data.



Index-Organized Tables

An index-organized table has a storage organization that is a variant of a primary B-tree. Unlike an ordinary (heap-organized) table whose data is stored as an unordered collection(heap), data for an index-organized table is stored in a B-tree index structure in a primary key sorted manner. The structure of an index-organized table provides the following benefits:

- Fast random access on the primary key because an index-only scan is sufficient.
- Fast range access on the primary key because the rows are clustered in primary key order.
- Lower storage requirements because duplication of primary keys is avoided. Index-organized tables have full table functionality. They support features such as constraints, triggers, LOB and object columns, partitioning, parallel operations, online reorganization, and replication. And, they offer these additional features:
- Key compression
- Overflow storage area and specific column placement
- Secondary indexes, including bit map indexes.

Creating an Index-Organized Table

```
CREATE TABLE admin_docindex(
    token char(20),
    doc_id NUMBER,
    token_frequency NUMBER,
```

Heap-organized table

- A heap-organized table is a table with rows stored in no particular order.
- This is a standard Oracle table; the term "heap“ is used to differentiate it from an index-organized table or external table.
- If a row is moved with in a heap-organized table, the row’s ROWID will also change.

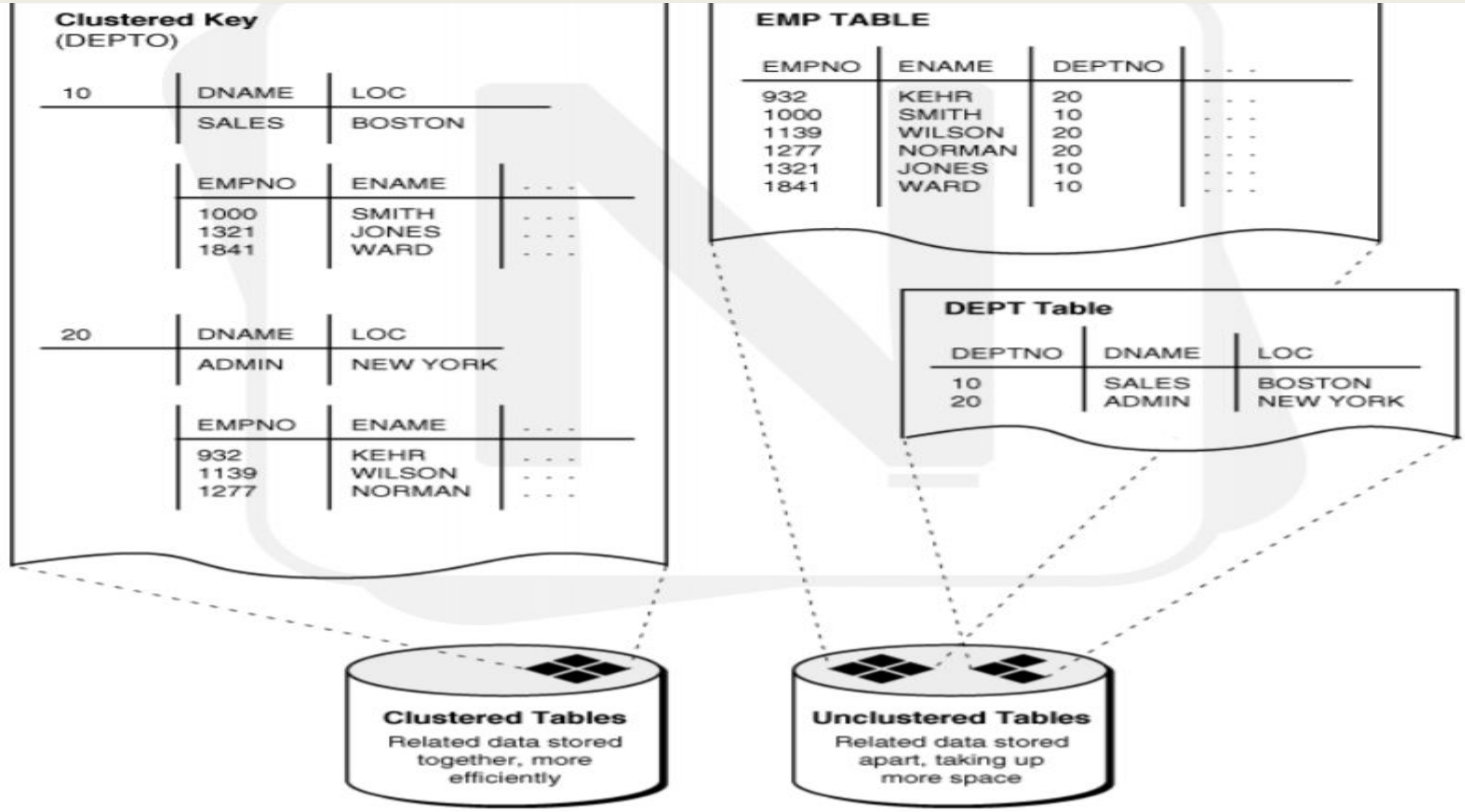
Examples

```
CREATE TABLE t1 (c1 NUMBER PRIMARY KEY, c2 VARCHAR2(30));
CREATE TABLE t1 (c1 NUMBER PRIMARY KEY, c2 VARCHAR2(30)) ORGANIZATION HEAP;
```

Clusters

- ➢ A cluster provides an optional method of storing table data.
- ➢ A cluster is made up of a group of tables that share the same data blocks.
- ➢ The tables are grouped together because they share common columns and are often used together.
- ➢ Because clusters store related rows of different tables together in the same data blocks, properly used clusters offer two primary benefits:

- Disk I/O is reduced and **access time** improves for joins of clustered tables.
- The cluster key is the column, or group of columns, that the clustered tables have in common. You specify the columns of the cluster key when creating the cluster. Therefore, **less storage** might be required to store related table and index data in a cluster than is necessary in non-clustered table format.



Cluster Types

- **Fail-Over Clusters**: emphasis is on complete avoidance of unplanned downtime and achieving higher availability. High-availability clusters allow multiple servers, in conjunction with shared disk storage units, to quickly recover from failures.
- **Scalable High Performance Clusters**: provides scalability, high performance, load balancing, and high availability through the use of parallel middleware. These are also called Parallel Clusters or High Performance Computing Clusters (HPCC).
- **Application Clusters:** Real Application Clusters (Oracle RAC) is a clustered version of Oracle Database based on a comprehensive high-availability stack that can be used as the foundation of a database cloud system as well as a shared infrastructure, ensuring high availability, scalability, and agility for any application.
- **Network Load balancing clusters**: Load balancing is the means by which network traffic directed to a particular Website is divided between one or more machines in a cluster of servers. System performance is optimized, scalability is simplified, and application availability (a key requirement of Web-based applications) is greatly enhanced.

MANAGING DATA AND CONCURRENCY

Manipulating data through sql

SQL language encompasses several categories of statements: statements that work with data, other statements that define and modify the structures (such as tables and indexes) that hold data, and still other statements that control the operation of the database itself.

Statements used to manipulate data:

- **INSERT**

Places new records, or rows, into a database table.

INSERT INTO table_name (column_list) VALUES (value_list);

INSERT INTO employee (employee_id, employee_name); VALUES ('114’,’Jasleen Bhatia’);

- **SELECT**

Retrieves previously inserted rows from a database table.

SELECT * FROM employee;

SELECT employee_id, employee_name, employee_hire_date, employee_termination_date, employee_billing_rate FROM employee;

SELECT employee_id, employee_name FROM employee WHERE employee_id=114;

- **UPDATE**

Modifies data in a table.

```
UPDATE project
SET project_name = 'Virtual Private Network’,
project_budget = 199999.95
WHERE project_id = 1005;
```

- **DELETE**

Deletes data from a table.

```
DELETE FROM project
WHERE project_id>8000;
TRUNCATE TABLE table_name;
```

- **MERGE**

Brings a table upto date by modifying or inserting rows.

Function

A function is a named PL/SQL Block which is similar to a procedure. The major difference between a procedure and a function is, a function must always return a value, but a procedure may or may not return a value.

Creating a Function

```
CREATE [OR REPLACE] FUNCTION function_name

[(parameter_name [IN|OUT|IN OUT] type[,...])]

RETURN return_datatype

BEGIN <function_body>

END[function_name];
```

Procedures

A procedure is a named PL/SQL Block which is similar to a function. The major difference between a procedure and a function is, a function must always return a value, but a procedure may or may not return a value.

Creating a Procedure

```
CREATE [OR REPLACE] PROCEDURE procedure_name
[(parameter_name[IN|OUT|IN OUT]type[,...])]
BEGIN
    <procedure_body>
END procedure_name;
```

Packages

PL/SQL packages are schema objects that groups logically related PL/SQL types, variables and subprograms.

A package will have two mandatory parts:

- Package specification: The specification is the interface to the package. It just DECLARES the types, variables, constants, exceptions, cursors, and subprograms that can be referenced from outside the package. In other words, it contains all information about the content of the package, but excludes the code for the subprograms.
- Package body or definition: The package body has the codes for various methods declared in the package specification and other private declarations, which are hidden from code outside the package

Triggers

Triggers are stored programs, which are automatically executed or fired when some events occur. Triggers are, in fact, written to be executed in response to any of the following events:

- A database manipulation (DML) statement (DELETE, INSERT, or UPDATE).
- A database definition (DDL) statement (CREATE, ALTER, or DROP).
- A database operation (SERVER ERROR, LOG ON, LOG OFF, START UP, or SHUTDOWN).

Triggers could be defined on the table, view, schema, or database with which the event is associated.

Locking Concept

- A lock is a mechanism that prevents destructive interactions, which are interactions that incorrectly update data or incorrectly alter underlying data structures, between transactions accessing shared data.
- Locks play a crucial role in maintaining database concurrency and consistency.
- In general, the database uses two types of locks: exclusive locks and share locks
- Only one exclusive lock can be obtained on a resource such as a row or a table, but many share locks can be obtained on a single resource.
- Locks affect the interaction of readers and writers.
- The following rules summarize the locking behavior:

- A row is locked only when modified by a writer.
- A writer of a row blocks a concurrent writer of the same row.
- A reader never blocks a writer.
- A writer never blocks a reader.

Lock	Description
DML locks (data locks)	DML locks protect data. For example, table locks lock entire tables, row locks lock selected rows.
DDL locks (dictionary locks)	DDL locks protect the structure of schema objects—for example, the definitions of tables and views
Internal locks and latches	Internal locks and latches protect internal database structures such as datafiles. Internal locks and latches are entirely automatic.

INTRODUCTION TO SPREAD SPECTRUM

Introduction

Application

Need

Significance

Block Diagram

Types

By Jasleen Kaur

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- **Problems encountered by a communication system** :

i. In area such as military the information has to be secured and should not be allowed to interfere communication by any means.

ii. Sometimes a hostile transmitter can jam transmission. To avoid this the channel should be immune to any external interference.

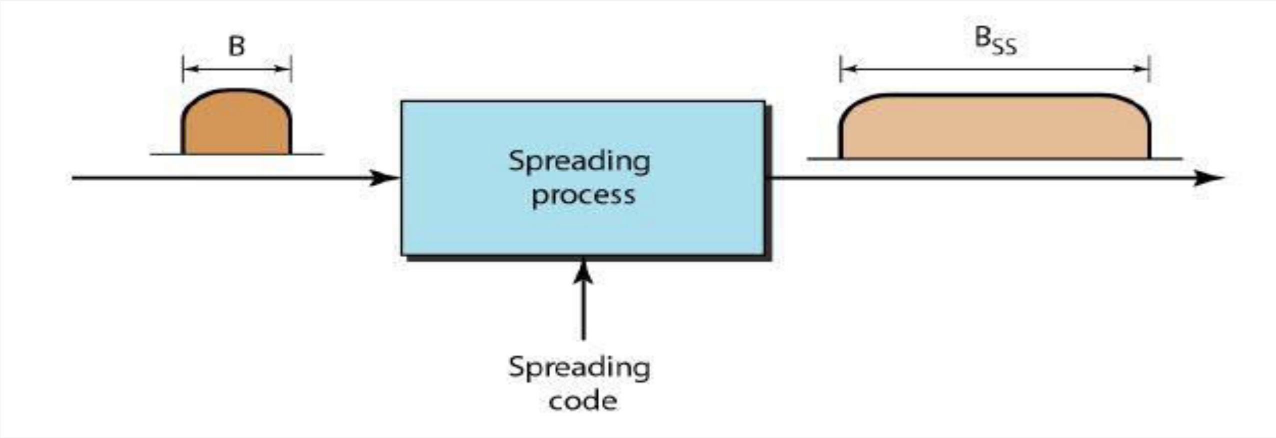
iii. Even in non military communications an unintentional interference is caused by a user who is transmitting its information through a channel which is already being used.

These problems can be solved using a technique called spread spectrum.

SPREAD SPECTRUM

- It is modulation technique in which transmitted BN is larger then information signal BN.
- Bit rate of spreading sequence is much larger than the input data.
- SS signal occupies a larger bandwidth than that of normal signal .
- Problem such as capacity limits, propagation effects, synchronization occur with wireless systems.
- Spread Spectrum modulation spreads out the modulated signal bandwidth so it is much greater than the message bandwidth.

- Bss>>B (BSS: Spreading Bandwidth, B:Bandwidth of signal)
- BSS=LB ;
- Processing Gain or Spreading Factor; L= BSS/B {for Analog System}
- L=Tb/Tc (Tb:bit time period , Tc: chip time period) {for Digital System}

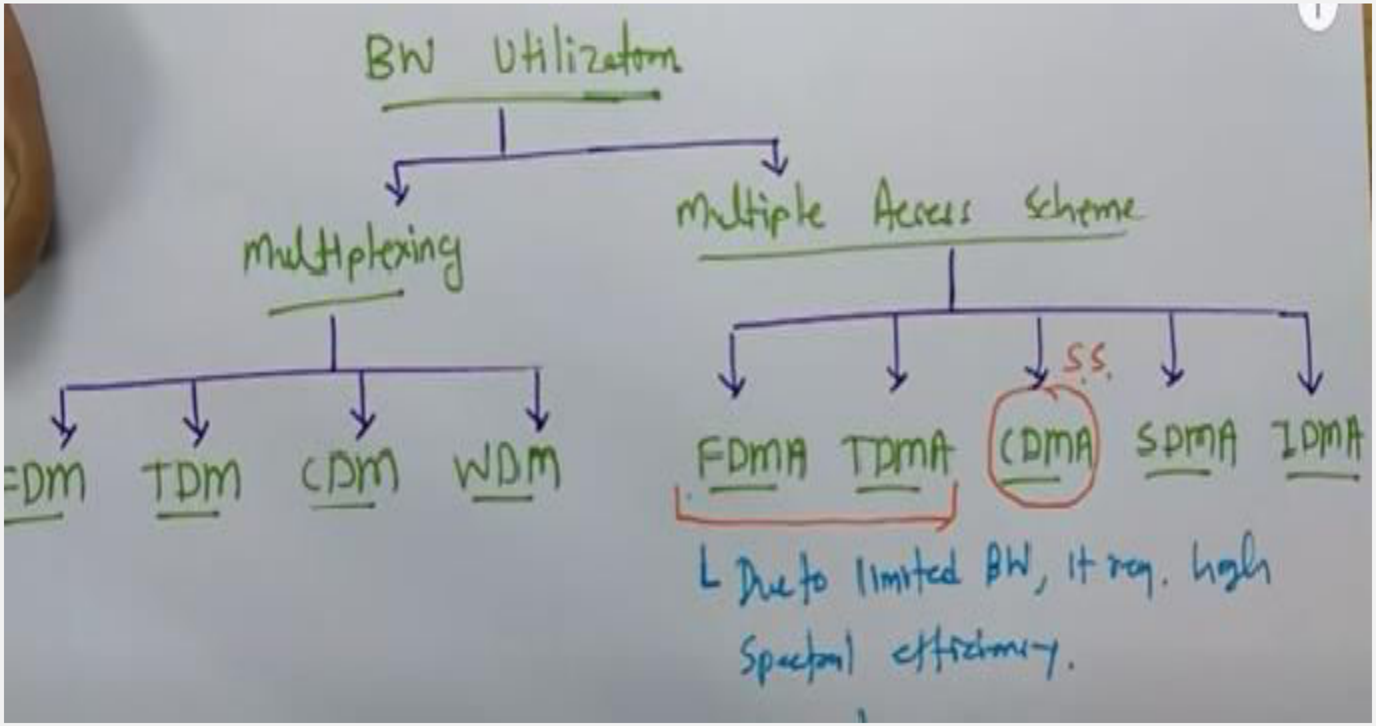


- Application of SS Modulation Technique:

i. In combating the intentional interference (jamming)
ii. In rejecting the unintentional interference from some other user.
iii. To avoid the self interference due to multipath propagation.
iv. In obtaining the message privacy.

- Need of Spread Spectrum :

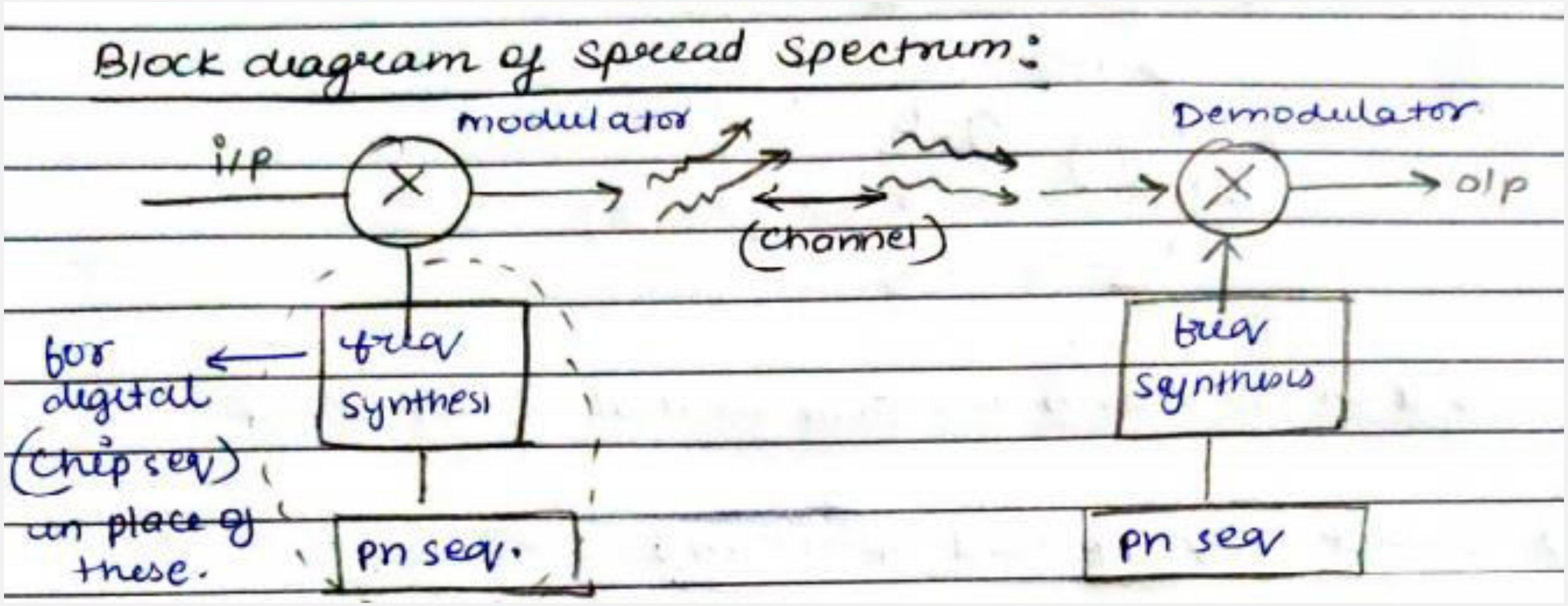
i. Improves the limitation of FDMA and TDMA which are:
   - Limited bandwidth , it require high spectral efficiency
   - Due to limited bandwidth , Concentrated Spectrum
   - Due to limited bandwidth , Narrowband Spectrum



BW Utilization
Multiplexing
Multiple Access Scheme
DM
TDM
CDM
WDM
FDMA
TDMA
S.S.
CDMA
SDMA
IDMA
Due to limited BW, it req. high
Spectral efficiency.

- Significance of Spread Spectrum

i. Immunity to jamming
ii. Low Interference
iii. High Processing Gain
iv. Easy Encryption
v. Greater Security
vi. Multiple Access



- A pseudo-noise (PN) sequence may be defined as a coded sequence of 1s & 0s with certain auto-correlation properties.
- These Additional frequency from frequency synthesisers is transmitted with the input data based on the data in PN sequence to spread the spectrum of data and reverse happens at the reception end.

- **There are two types of spread spectrum systems:**

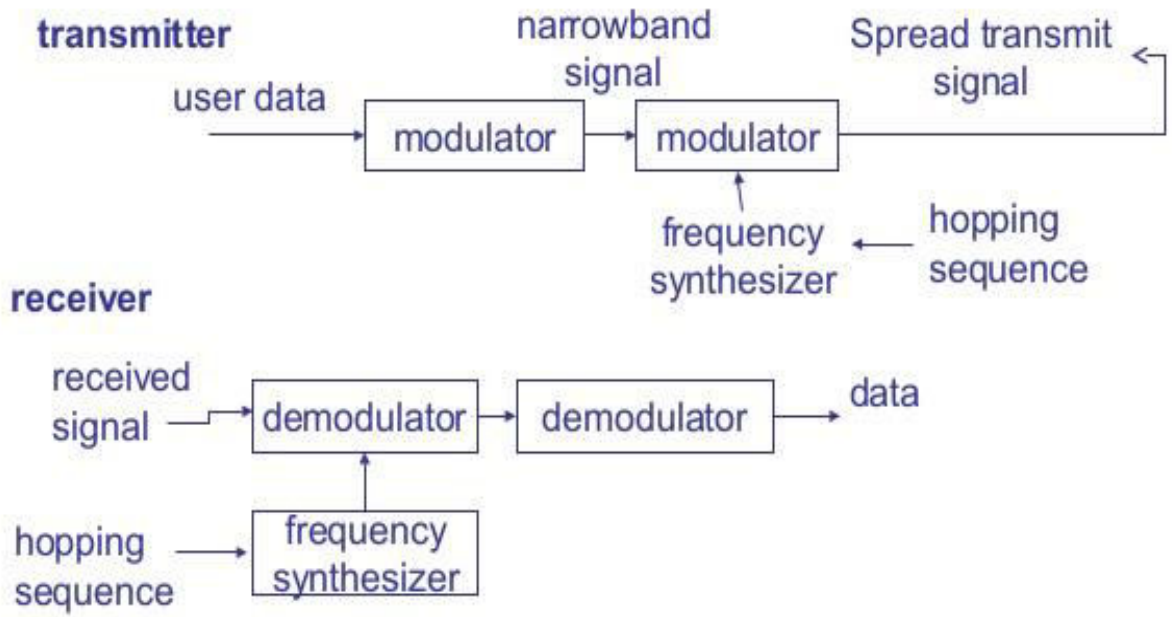
1) **Frequency Hopping Spread Spectrum (FH-SS):** In this, the data is used to modulate carrier. The data modulated carrier is then randomly hoped from one frequency to the other. Because of this , spectrum of transmitted signal is spreaded sequentially rather than instantaneously.

- Uses MFSK (M-ary Frequency Shift Keying) Modulation technique.
- Two types:

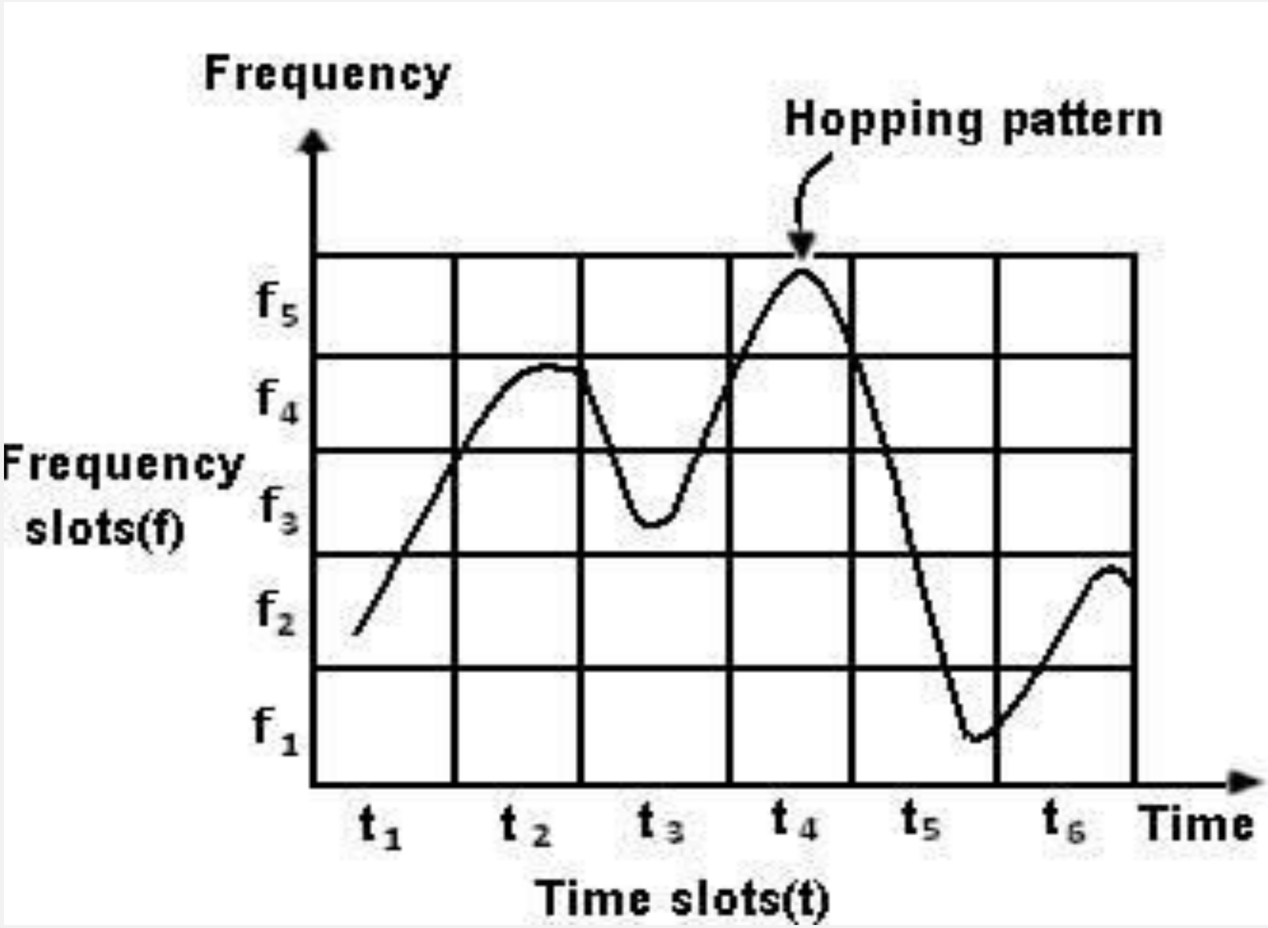
i. **Slow frequency hopping:** In this, the symbol rate (Rs) of the MFSK signal = integer multiple of the hop rate (Rh): i.e. each frequency hop=several symbols.

ii. **Fast frequency hopping:** In this, the hop rate (Rh) = integer multiple of symbol rate of MFSK (Rs). Therefore, each symbol transmission=several frequency hops.

FHSS (Frequency Hopping Spread Spectrum)



Block Diagram

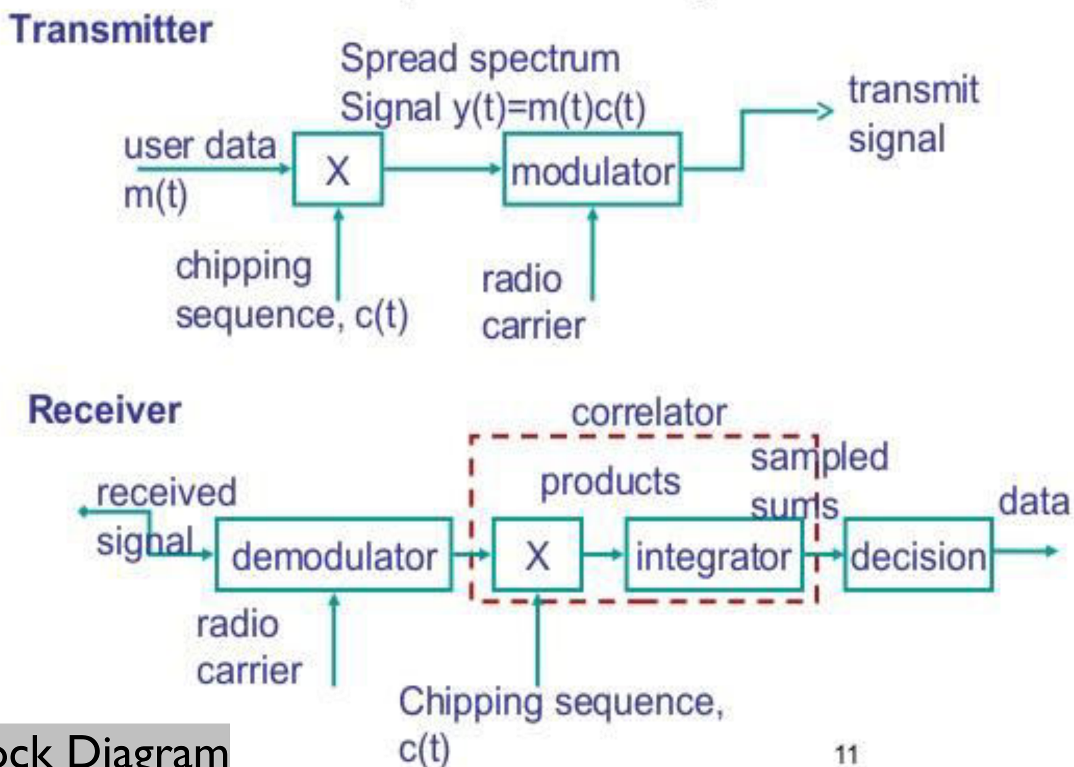


Frequency
Hopping pattern
Frequency slots(f)
f5
f4
f3
f2
f1
t1
t2
t3
t4
t5
t6
Time
Time slots(t)

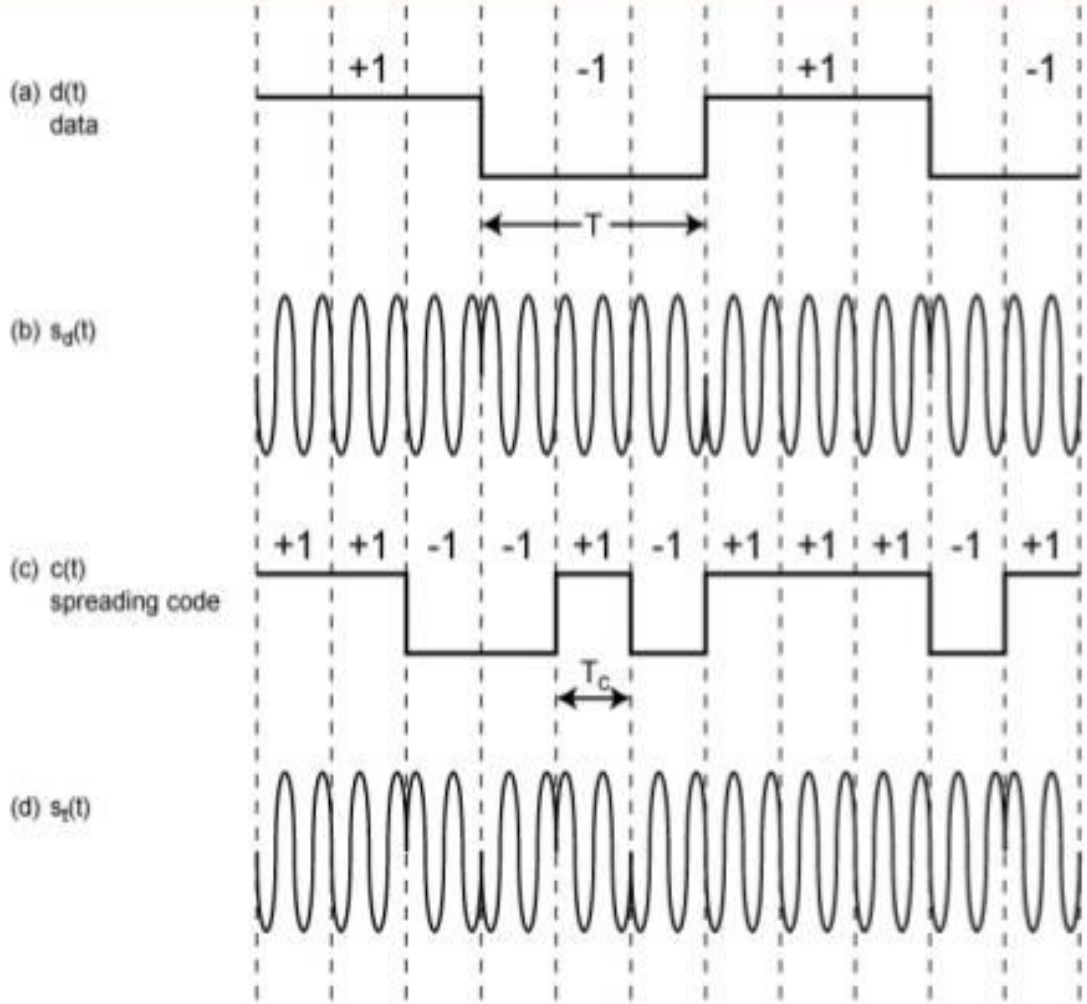
2) Direct Sequence Spread Spectrum(DS-SS): In this, can be used in practice for transmission of signal over a bandpass channel (ex-satellite channel)

- Uses two major signaling techniques PSK (phase shift keying) DSSS & FSK (frequency shift key) FHSS.
- Every user assigned a spreading code. This secret code is used to encode he signal.
- This code is multiplied with the original message and resultant message s than transmitted.
- Receiver use same spreading code to decode the message o retrieve original message.

DSSS (Direct Sequence Spread Spectrum)



Block Diagram


(a) d(t)
data
+1
-1
+1
-1
T
(b) $s_d(t)$
(c) c(t)
spreading code
+1 +1 -1 -1 +1 -1 +1 +1 +1 -1 +1
T_c
(d) $s_t(t)$

UNIT-2 2G NETWORKS

Introduction

GSM

GSM Services

Architecture

GSM Interface

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INTRODUCTION

- The second generation (2G) cellular systems provide more facilities and attractive features than first generation (1G) systems.
- The 2G standards rely on digital formats including TDMA/FDD and CDMA/FDD.
- 2G standards provide:

i. Better speech quality

ii. High speed data application

iii. Efficient spectrum usage and compatible with TDMA access scheme.

iv. Support multiple users.

- The TDMA technique is used by 2G standards that include Global system for mobile (GSM). The TDMA schemes use half duplex method. Other TDMA feature include:

i. Good Synchronization
ii. For better operations, it possesses guard intervals.
iii. Efficient spectrum utilization.
iv. Operates with faster data rates.

- With CDMA technique, every subscriber is assigned a n-bit code and it guarantees high degree of security than other access techniques.
- Thus, TDMA and CDMA in 2G standards provide less design complexity, more security, faster speeds of transmissions and high reliability than 1G systems.

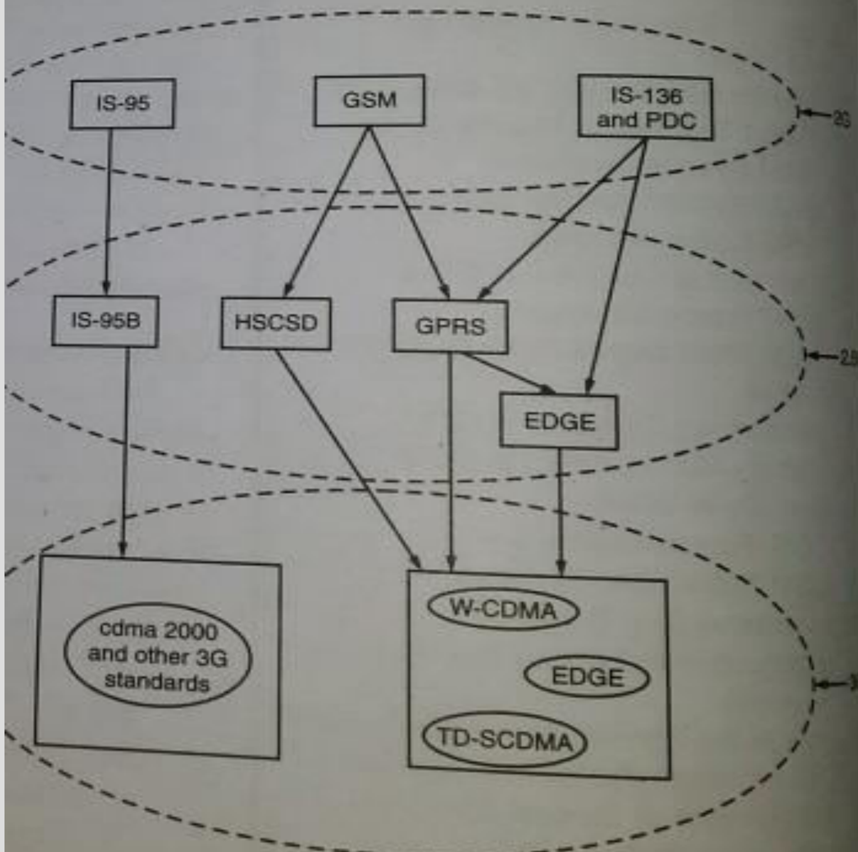


Fig. 4.2. Some of the 2G to 3G technology mobile standards-path upgrade view

➢ IS-136 (D-AMPS):

1) It stands for Interim standard-136.
2) The analog mobile phone system (AMPS) was not suitable to support high traffic and demands for high capacity.
3) In 1980's USDC came into existence to accommodate multiuser with frequency spectrum allocated.
4) USDC/AMPS was standardized as IS-54, It allows dual mode systems.

✧ Salient Feature of USDC:

i. The USDC system uses frequency reuse technique.
ii. It provides high cellular capacity.
iii. It accommodate many users.
iv. It was also called North America Digital Cellular Standard(NADC).
v. USDC shares same frequencies with minimized interference.

✧**Salient Feature of IS-136**

i. It operates in 800-1900 MHz frequency band.

ii. Its channel bandwidth is 30kHz.

iii. The TDMA frame with 40 msec in 6 time slots is being used in it.

iv. Six time slots: (channel data rates)

1st time slot –Half rate channel

2nd , 5th time slots- Full rate channel

2nd,3rd, 5th ,6th time slots- Double rate channel

v. The data link layer of IS-136 provides,

a) Addressing

b) Flow Control

c) Error detection

d) Segmentation

e) Media Access Control

vi) Network layer allows:

a) Establishments

b) Maintaining

c) Termination of connection

vii) It possesses channels namely,

a) Digital Control Channel

b) Digital Traffic channel

➢IS-95 (CDMA):

1) Interim standard 95.
2) It can tolerate interference with spread spectrum technique.
3) It provides high security.
4) It helps in achieving high capacity land mobile communication systems.

✧Salient Feature of IS-95:

i. It uses speech coder Qualcomm, code Excited Linear Predictive Coder, which can detect voice activities & reduce data rates upto 1200 b/sec in silent period.
ii. It permits every subscriber within a cell to make use of radio channel where same radio channel is used by subscribers of adjacent cells too. This is possible with direct sequence spread spectrum (DSSS) technique.

iii. It uses specific modulation and spread spectrum techniques in its forward as well as reverse links.

iv. It is compatible with IS-41 networking standard.

➢ **Channels and frequencies used in IS-95:**

a. Forward Link: 869-894 MHz

b. Reverse link: 824-849MHz

c. For cellular operation, channel pair separated by frequency of 45MHz.

d. Total Spreading factor is 128 (chip rate of 1.228 M chips/sec)

- **GSM (Global System for Mobile)**:
- The main aim was to provide ‘roaming facility’ for subscribers & to follow ISDN guidelines.
- Its is first 2G cellular network.
- 2G network developed as a replacement for first generation (1G) analog cellular networks.
- The GSM standard originally described a digital, circuit –switched network optimized for full duplex voice telephony.
- **Basic Requirement:**

i. Service Portability
ii. Quality of Service
iii. Frequency utilization (Reuse)
iv. Cost
v. Security

- **Performance characteristic of GSM**:

i. Communication: wireless digital communication; support for voice and data services.

ii. Total mobility: international access

iii. Worldwide connectivity: The network can handles localization

iv. High Capacity: Better frequency efficiency

v. High Transmission Quality: high audio quality, uninterrupted phone calls at higher speeds, Better handoffs

- **Disadvantages:**

i. High complexity of the system

ii. Several incompatibilities within the GSM standards

iii. No end to end encryption of data.

- **GSM Service:**

GSM services follow ISDN guidelines and are classified as either Teleservices or Data services. Teleservices include standard mobile telephony and mobile originated or base originated traffic. Data Services include computer- to computer communication and packet switched traffic. User services divided into three categories:

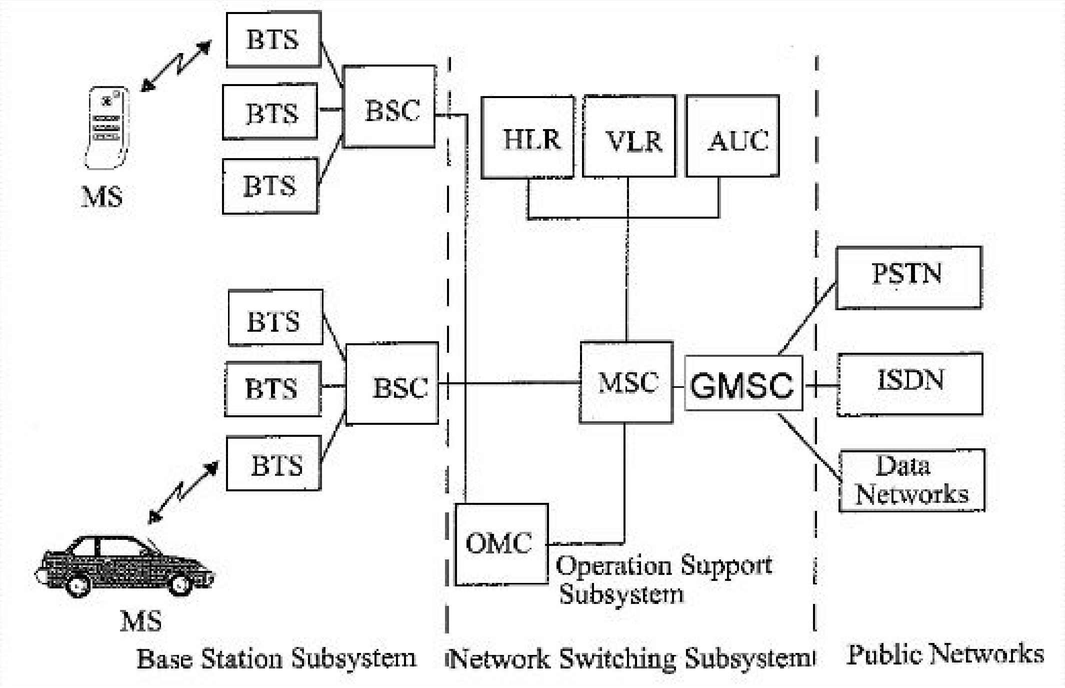
1) **Bearer Services** (Related to1,2 and 3 layer of OSI, Data service 300bps to 9.6kbps.)
2) **Telephone Services** (voice transmission, Message service and data communication, Emergency calls, SMS, MMS)
3) **Supplementary Services** (call forwarding, call redirecting)

- **GSM Architecture:**

The GSM system architecture consists of three major interconnected subsystems that interact with between themselves and with the users through certain network interfaces. The subsystems are the Base station Subsystem (BSS), Network and switching Subsystem(NSS) and the Operation Support Subsystem(OSS). The mobile station(MS) is also a subsystem, but is usually considered to be part of the BSS for architecture purposes.

Global System for Mobile Communications

Basic architecture



1) **Base Station Subsystem(BSS)**: The BSS provides and manages radio transmission path between the Mobile Station(MS) and the mobile Switching Centre(MSC). It also manages radio interface between the mobile stations and all other subsystems.

i. Mobile Stations(MS): The MS communicates the user information and modifies it according to the transmission protocols of air-interface to communicate with other BSS equipment. The MS has two elements, Mobile equipment(ME) and Subscriber Identity Module (SIM).

ii. Two Elements in BSS: Base Transceiver Station(BTS) & Base Station Controller(BSC)

- Base Transceiver Station(BTS): The BTS include a transmitter, a receiver and signaling equipment to operate over the air interface. It include all necessary equipment for radio transmission
- Base Station Controller: It manages BTS. It manages handoffs, radio frequency assignment

2) **Network and switching Subsystems(NSS):**

The NSS handles the switching of GSM calls between external networks and the BSC's in the radio subsystem and is also responsible for managing and providing external access to several customer databases. It provides the main control and interfacing for whole network.

i. Mobile Switching Centre(MSC): It is the main element within the core of GSM network. The MSC is the central unit in the NSS and controls the traffic among all of the BSCs.

ii. Three Databases:

- HLR(Home Location Register): Contains all the administrative information about subscriber and location information for each user who resides in the same city.
- VLR(Visitor Location Register): VLR database which temporarily stores the customer information for each roaming subscriber.
- Authentication Centre(AUC): Handles authentication and encryption keys for every single subscriber in the HLR and VLR.

iii. **Operation Support Subsystem(OSS)**: The OSS supports one or several Operation Maintenance Centre (OMC) which used to monitor and maintain the performance of each MS,BS,BSC and MSC within a GSM system. The OSS has three main functions, which are:

a) Manage all charging and billing procedures.

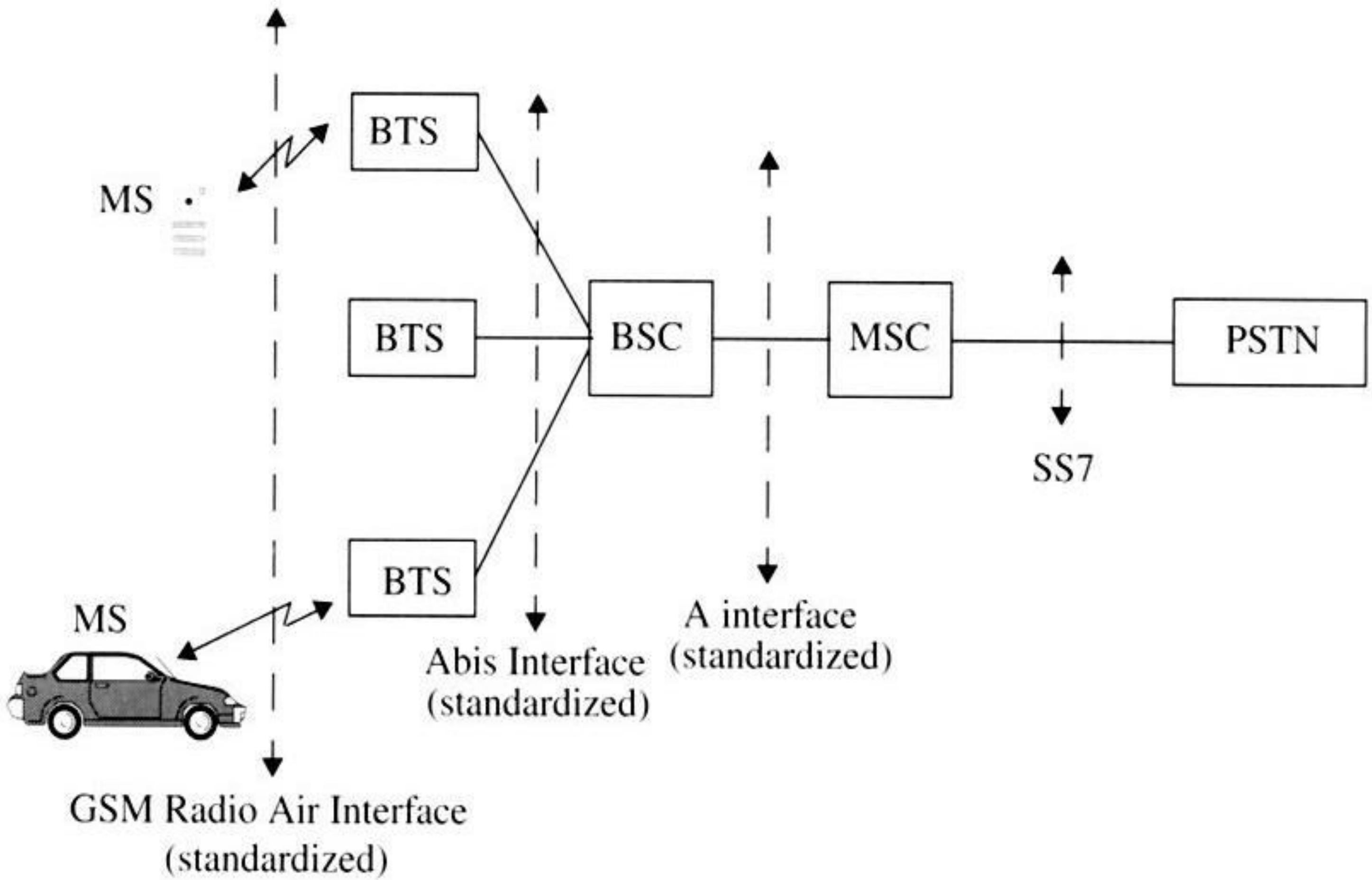
b) Manage all mobile equipment in the system.

c) To maintain all Telecommunications hardware and network operation.

3) **GSM interfaces**:

- The MS communicates with the Base Transceiver Station(BTS) over the Radio-Air Interface and the interface which connects BTS to a Base Station Controller(BSC) is called the Abis Interfaces.
- The Abis interface carries traffic and maintenance data.
- The interface between a BSC and a MSC is called the A interface. The A interface uses an SS7 protocol called the signaling correction control part(SCCP) which supports communication between the MSC and the BSS, as well as network messages between the individual subscribers and the MSC.
- The A interface allows a service provider to use base stations and switching equipment made by different manufacturers.

GSM interface



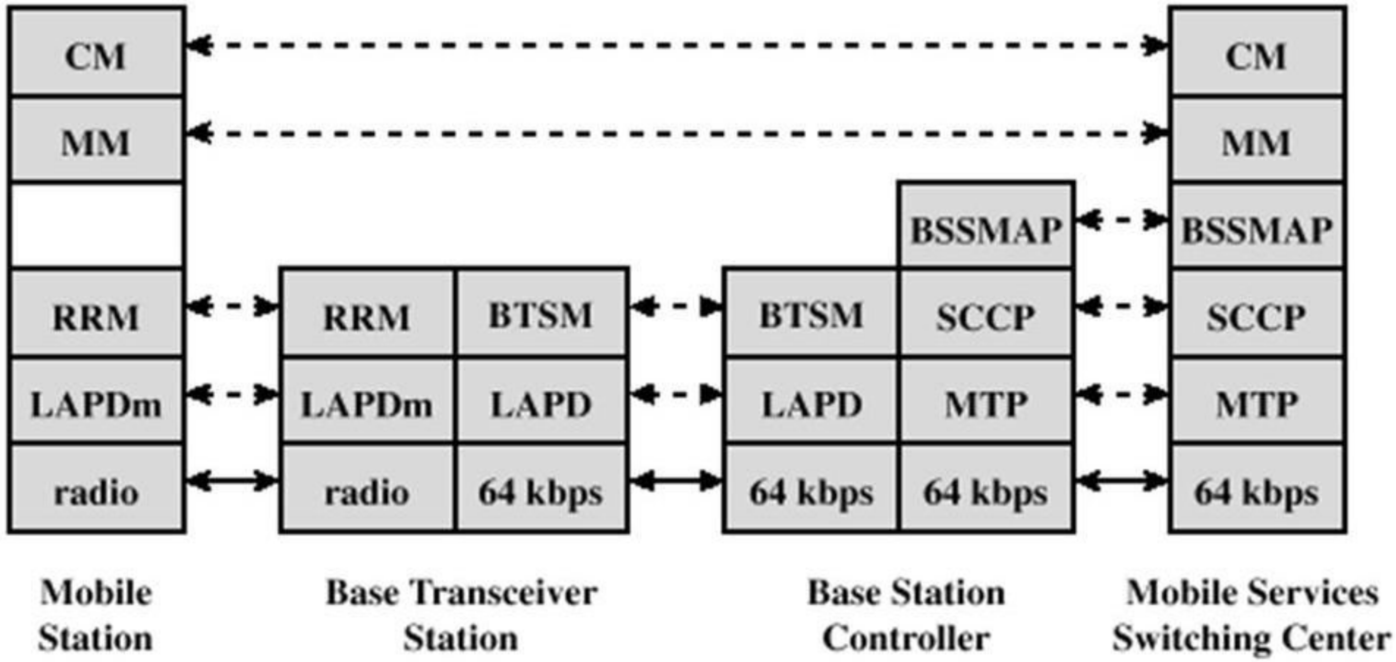
NETWORK SIGNALING IN GSM

- GSM signaling defines the communication between the mobile and the network.
- Signaling has to be carried through the network and across the air interface to the mobile.
- Different protocols are used across different interfaces.
- All GSM signaling is based on Open System Interconnect(OSI) model used for computer system.
- The signaling protocol in GSM is structured into three layers:

➢ Layer 1:Physical Layer- is providing media for communication

➢ Layer2: Data Link Layer: Error Detection & Correction

➢ Layer 3: Network Layer: Communication



BSSMAP	=	BSS mobile application part	MM	=	mobility management
BTSM	=	BTS management	MTP	=	message transfer part
CM	=	connection management	RRM	=	radio resources management
LAPD	=	link access protocol, D channel	SCCP	=	signal connection control part

Figure 10.17 GSM Signaling Protocol Architecture

- MS-BTS: The physical layer between MS and BTS is called Um or air interface and perform functions full or half duplex access, provides TDMA, FDMA & CDMA, framing of data.
- The Data link layer control the flow of packets to and from network and provides access to various services like:

i. Connection: provides connection between two terminal
ii. Teleservices: MMS,SMS, fax etc.

- The data link present between MS & BTS is called LAPDm and functions are data flow control, acknowledged/ unacknowledged data transmission, segmentation , address and sequence number check.
- CM: Connection Management

i. Supports call establishment, maintenance, termination
ii. Support Functioning of SMS
iii. Support Dual Tone Multiple frequency (DTMF) signaling

- MM (Mobility Management):

i. Control the issue regarding mobility management, location updating & registration.

- RRM (Radio Resource Management)

i. Manages radio resources such as: frequency assignment, signal measurement.

- BTS-BSC signaling protocols:

The physical layer between BTS and BSC is called Abis interface, where, voice is coded by using 64 kbps. The connection between BTS and BSC is through wired network. The data link layer is LAPDm. Network layer protocol is BTS management which interact with BSSAP.

- BSC-MSC Signaling protocols:

Physical layer between BSC and MSC is called A interface. Data link layer protocol between BSC and MSC is MTP(Message Transfer Protocol) and SCCP (Signaling Connection Control Protocol). MTP and SCCP are part of SS7. Network layer protocols are the MSC are CM, MM and BSSAP.

- Mobility Management in GSM:

1. Mobility management function handles the function that arises due to mobility of the subscriber.
2. Main objective of MM is location tracking & call setup.
3. Registration Process of MS moving from one VLR to another VLR.

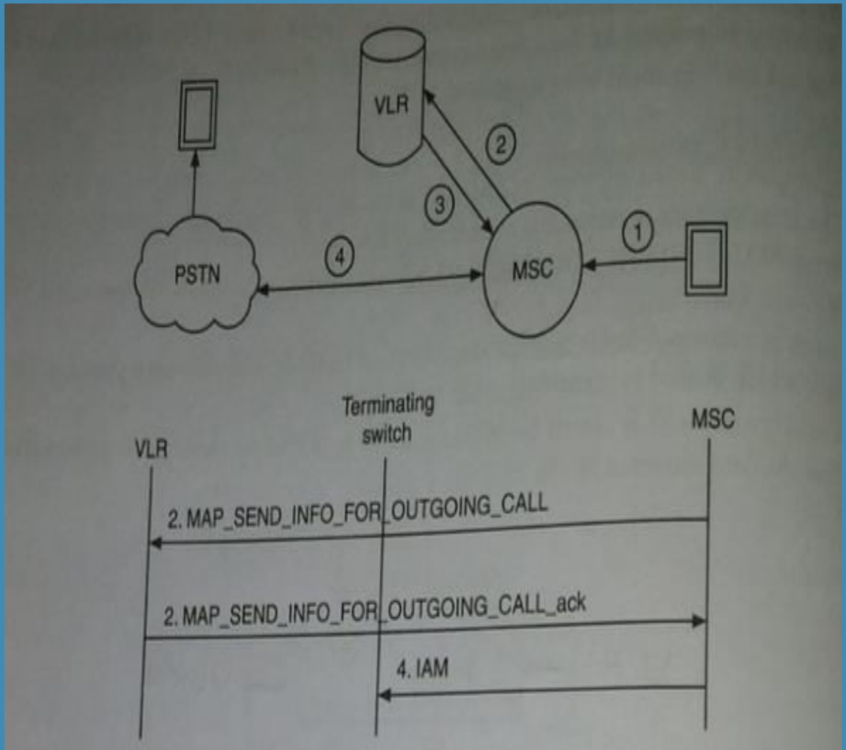


Fig. 4.29. GSM Call Origination Operation

GSM Call organization

i. MS sends the call organization request to MSC.

ii. MSC forwards the request to VLR by sending MAP_SEND_INFO_FOR_OUTGOING_CALL.

iii. VLR checks MS's profile and sends an ACK to MSC to grant call request.

iv. MSC set up communication link according to standard PSTN call setup procedure.

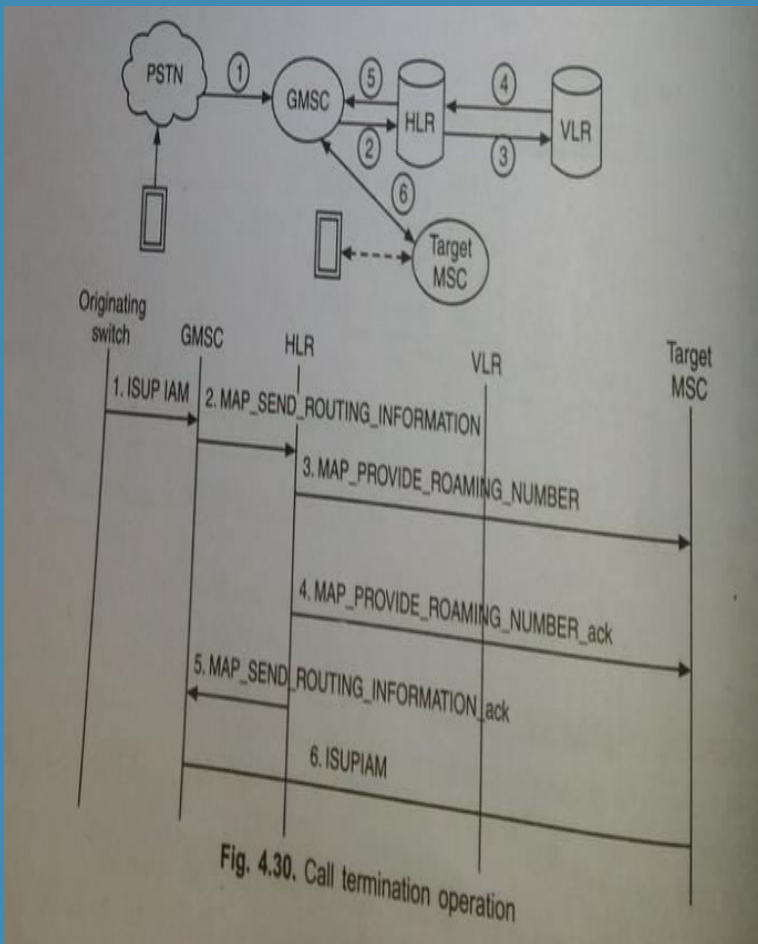


Fig. 4.30. Call termination operation

➢ GSM Call Termination

i. To obtain routing information, GMSC interrogates HLR by sending MAP_SEND_ROUTING_INFORMATION to HLR.

ii. HLR sends a MAP_PROVIDE_ROAMING_NUMBER msg to VLR to obtain MSRN (MS Roaming Number). The message consists of IMSI, MSC number etc.

iii. The VLR creates the MSRN by using MSC number stored in VLR record of MS. The MSRN number is sent back to GMSC through HLR.

iv. MSRN provides address of target MSC where the MS resides. Then a message is directed from GMSC to target MSC to set communication link.